WHERE THE PROJECT STANDS

Six stages since the last talk, one established recipe, one open mechanism

beetle now passes both shot counts

SAM 2's fine-tuning program is closed

Major's residual is not about the model

S15, S16, S17, S17b, S17c, S17d / S18 built and waiting / Hunter Bridges

RECAP

Where that talk left it

01 Seven techniques measured on SAM 2. The ones that paid all changed what the model is shown, not what it optimises.

02 The model swap paid on beetle: SAM 3 plus the architecture-adapted arm took the best 1-shot number from 64.12 to 69.88.

03 Every loss arm on SAM 2 was null, and nobody could say why.

04 Butterfly Major's residual survived curation, both rotation fixes, 5-shot, every loss and two model families.

Three questions were open: is a second labelled specimen worth more than a rotated one, what does the champion do at five shots, and why is every SAM 2 loss arm null.

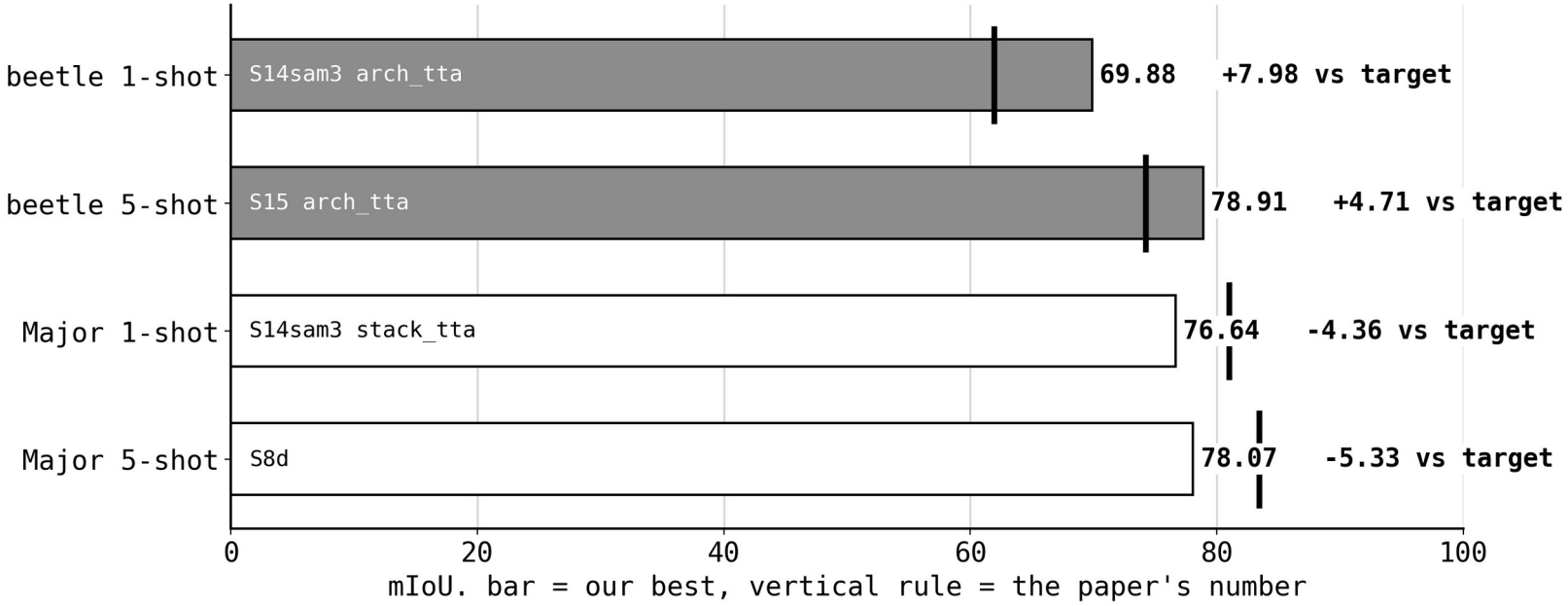
SINCE THEN

What landed since that talk

S16	which second reference pays	**PASS 14/14**
S15	the champion at five shots	**PASS 18/18**
S17	what the fine-tune does to its own pairs	**PASS 20/20**
S17b	the repairs, and whether they compose	**PASS 20/20**
S17c	the rest of the branch, in one session	**PASS 11/11**
S17d	the last cell, and the sentinel itself	**PASS 11/11**
S18	what may cross the centre of the cycle	**built, not yet run**

WHERE THE NUMBERS ARE

The scoreboard, as of today



Beetle reproduces at both shot counts and now exceeds the paper at both. Butterfly Major does not reproduce at either.

Everything in this talk is either one of those four numbers moving, or the reason one of them will not.

S16

Two ways to spend a second reference

IMPLICIT (orientation TTA)




the same specimen turned 180 degrees, outside the memory bank, picked per episode by a confidence rule

cost: 0 labels

EXPLICIT (a real 2-shot)



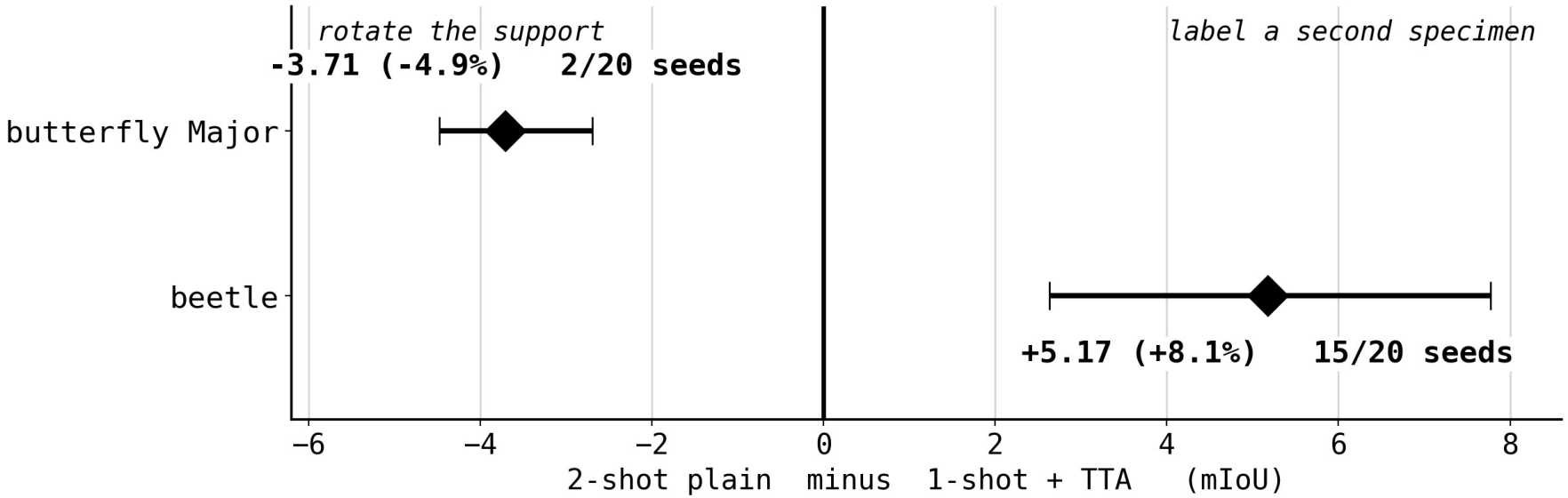

a second individual from the same pool, inside the bank, selected by memory attention

cost: 1 label

Every headline arm since S5c has used the implicit version. The explicit one had never been measured.

S16 RESULT

Which one pays flips by dataset

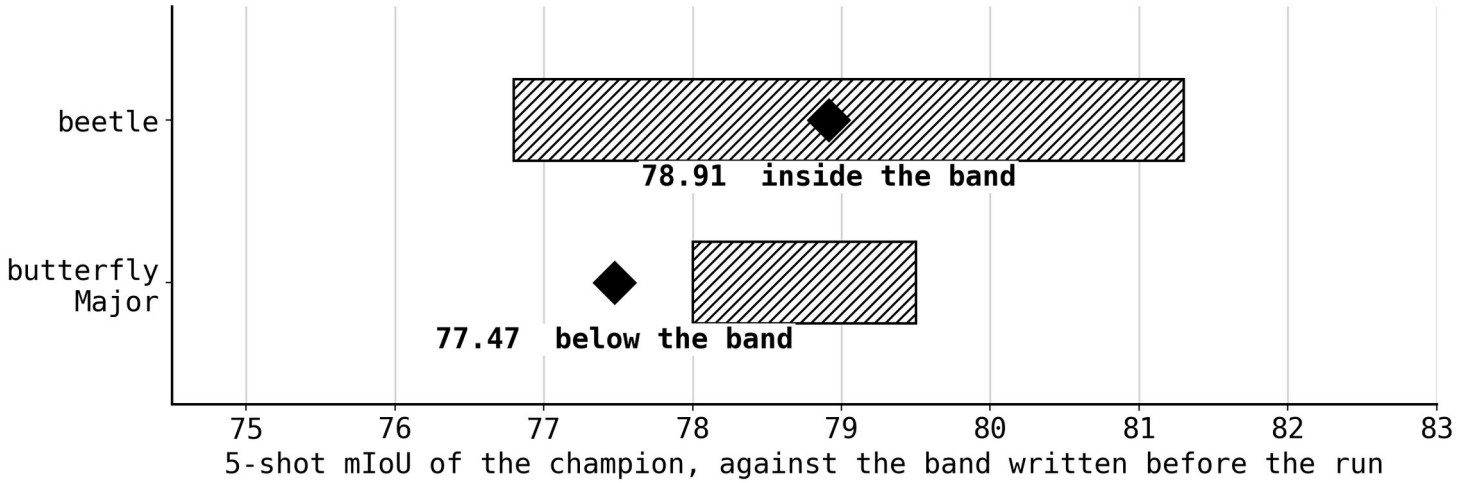


Both signs were pre-registered, from each dataset's own residual: Major's defect is query orientation, beetle's is species diversity, and a rotated copy adds no species.

The practical read: two labels on stock SAM 2 puts beetle at 69.04, within noise of the trained champion at one label.

S15

The champion at five shots

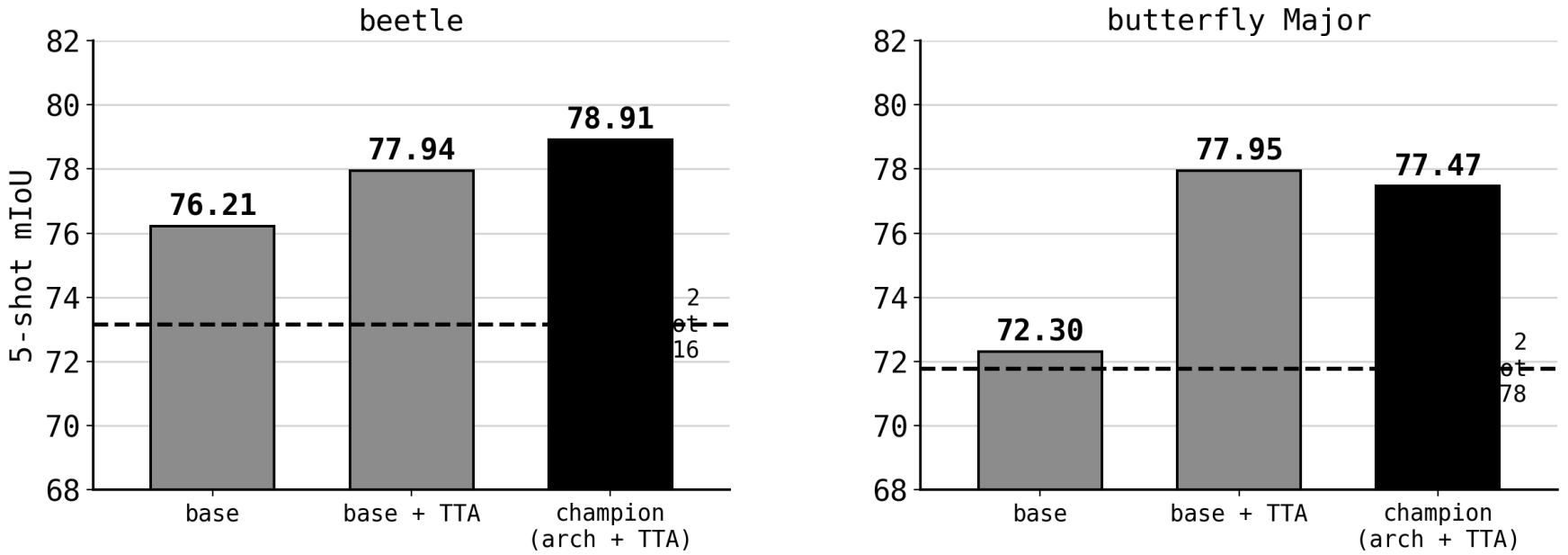


Bands written before the run, from two independent anchors: the silhouette score and the 1-shot oracle ceiling plus the measured shot mechanism.

beetle 78.91, inside. Gate PASS 18 of 18, and +3.3 above the paper's best beetle row.

Major 77.47, below its band, and beaten by its own base.

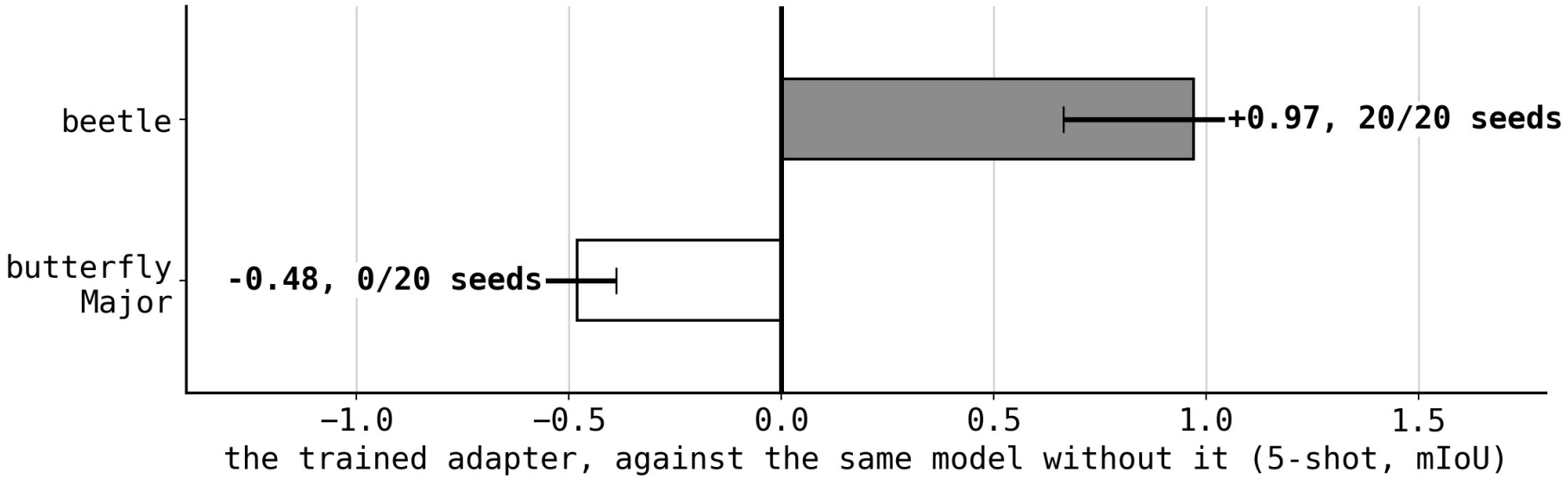
Every cell, both datasets



On beetle the adapter is the top of the stack. On Major the honest number is 77.95 with no adapter at all.

The backbone edge is real on both: SAM 3's plain weights beat SAM 2's by +3.06 on beetle and +0.52 on Major.

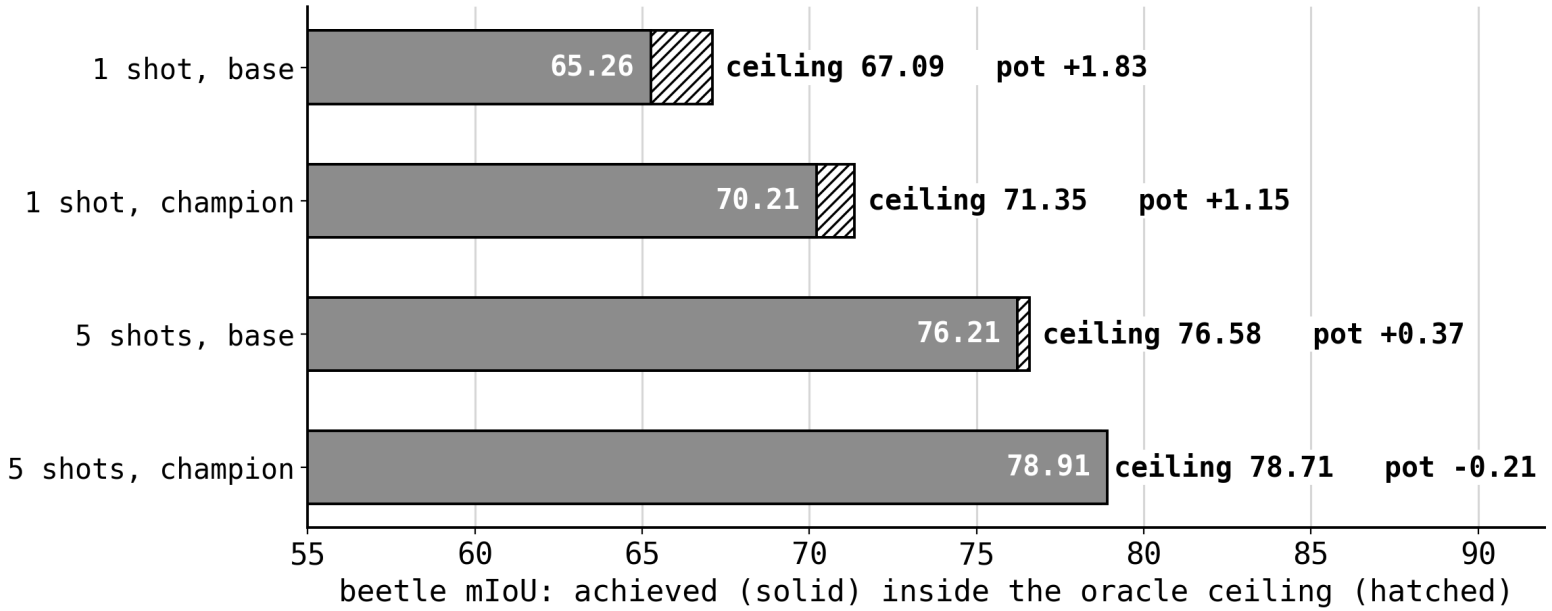
The adapter is beetle-specific



Same adapter, same protocol, same seeds. On beetle it pays on every seed. On Major it costs on every seed.

At 1 shot the beetle contribution was +1.96. At 5 shots it is +0.97, and the next slide is why.

Five labels close the pot

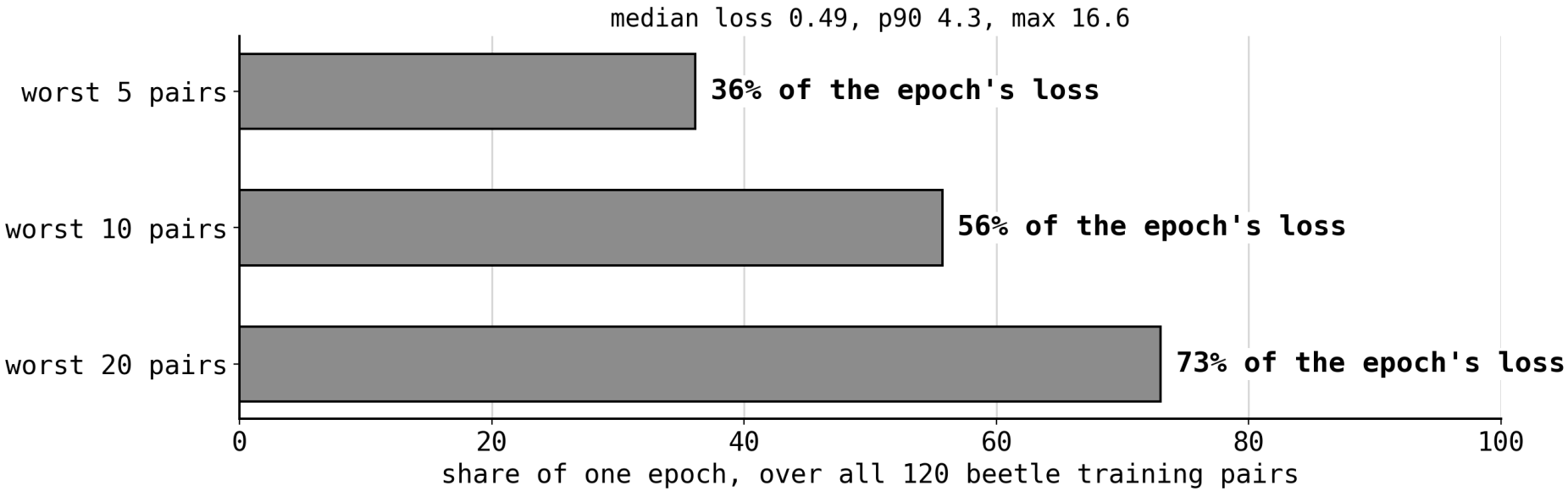


The hatched bar is what the same predictions would score if every proposed region were assigned to the right trait.

At five supports that gap is gone, so there is nothing left for an assignment loss to win. The pot is a function of the label budget, not of the objective.

S17 ANATOMY

One epoch is ten bad pairs

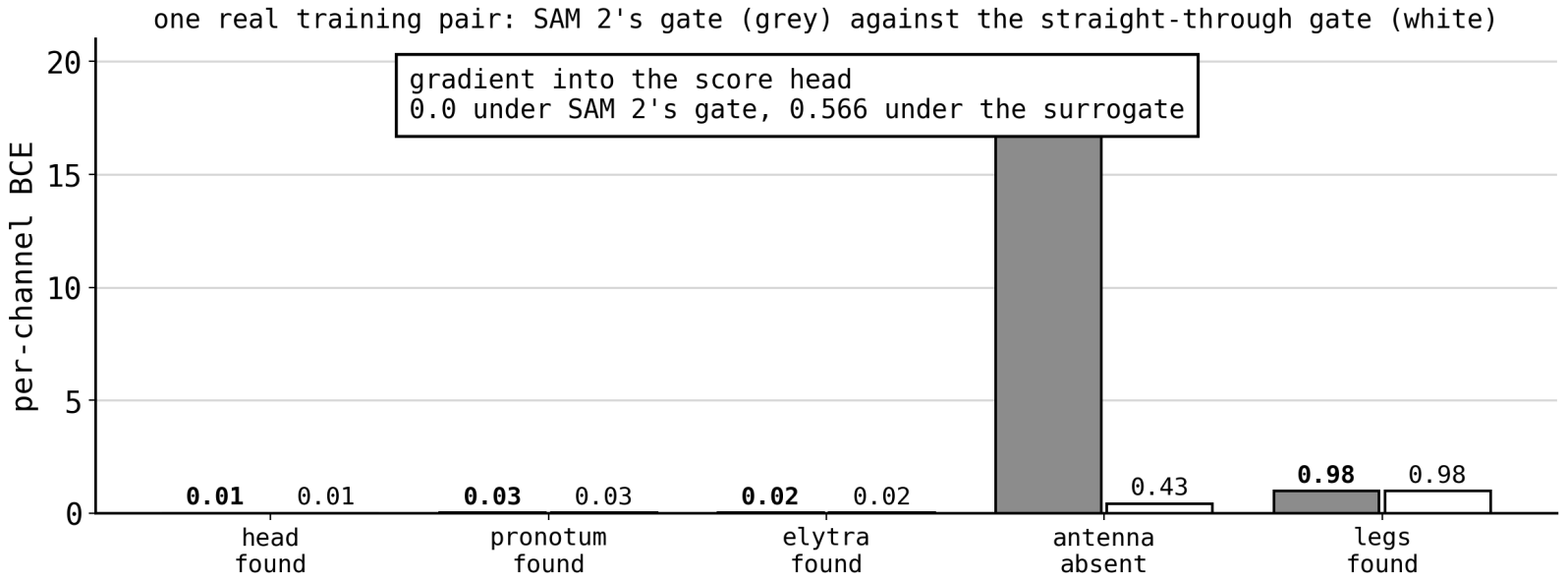


Measured on the base model over all 120 beetle training pairs, before any arm was designed.

The merged-silhouette recipe has no such tail. So the question became what those ten pairs are charging for.

S17 ANATOMY

What they are charging for

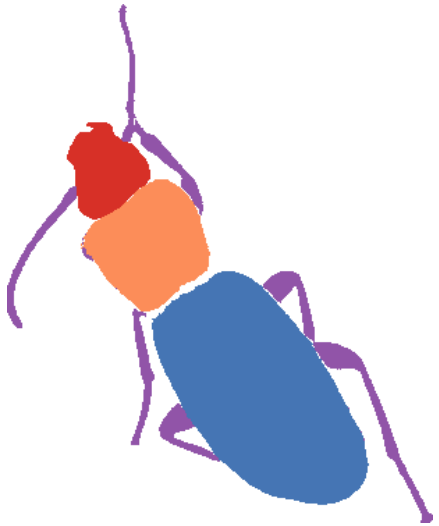


When SAM 2 decides a trait is absent it replaces the mask with a constant of -1024. Per-trait BCE then charges 1024 times the trait's area, on all 20 of the worst channels.

62% of the loss value is that constant, and it has no gradient. Nothing in the objective can reach a channel the model did not find.

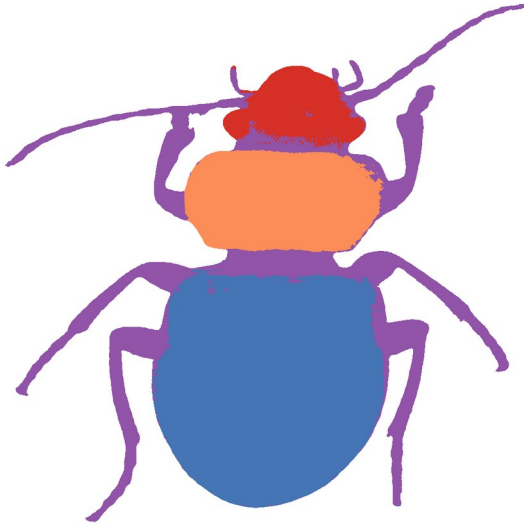
The cycle's own images

forward leg, on the query



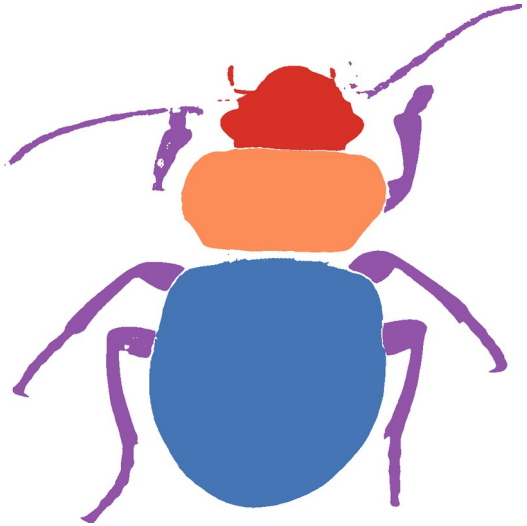
base 68.19, trained 67.50

closing leg, back on the reference



base 60.67

closing leg, trained arm



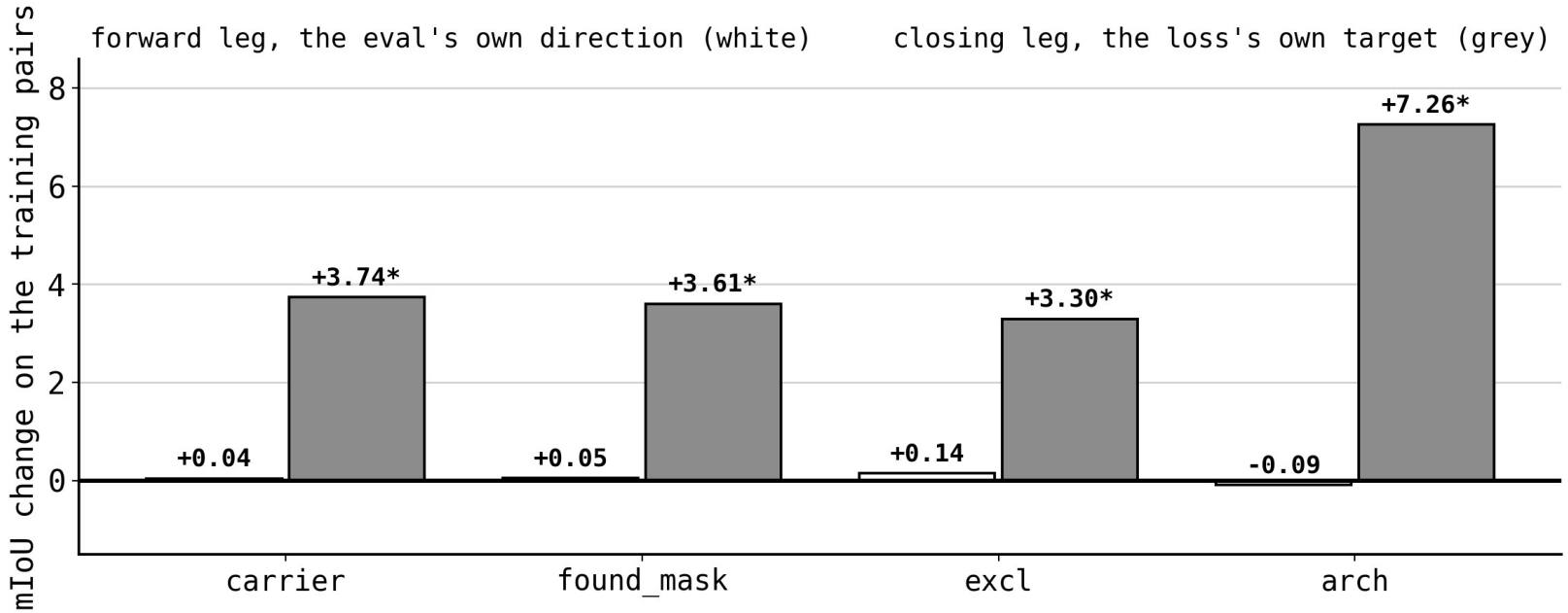
66.57

One real training pair, from the bundle. The antennae are missing from the forward prediction and taken over by the leg channel in the closing one.

Training moves the closing image. It does not move the forward one, which is the image the evaluator scores.

S17 RESULT

The reverse leg absorbs it all



Every arm lowers its own objective, and every arm does it on the closing leg. The correlation between the two legs is 0.01 to 0.31.

On butterfly the forward leg does move, +1.39, which is why OC-CCL pays there and not on beetle. The asymmetry is a property of the training pairs.

S17B

Three faults, three repairs

01 The closing head absorbs the objective. Repair: put the adapters on the forward legs only, so the closing frame cannot learn.

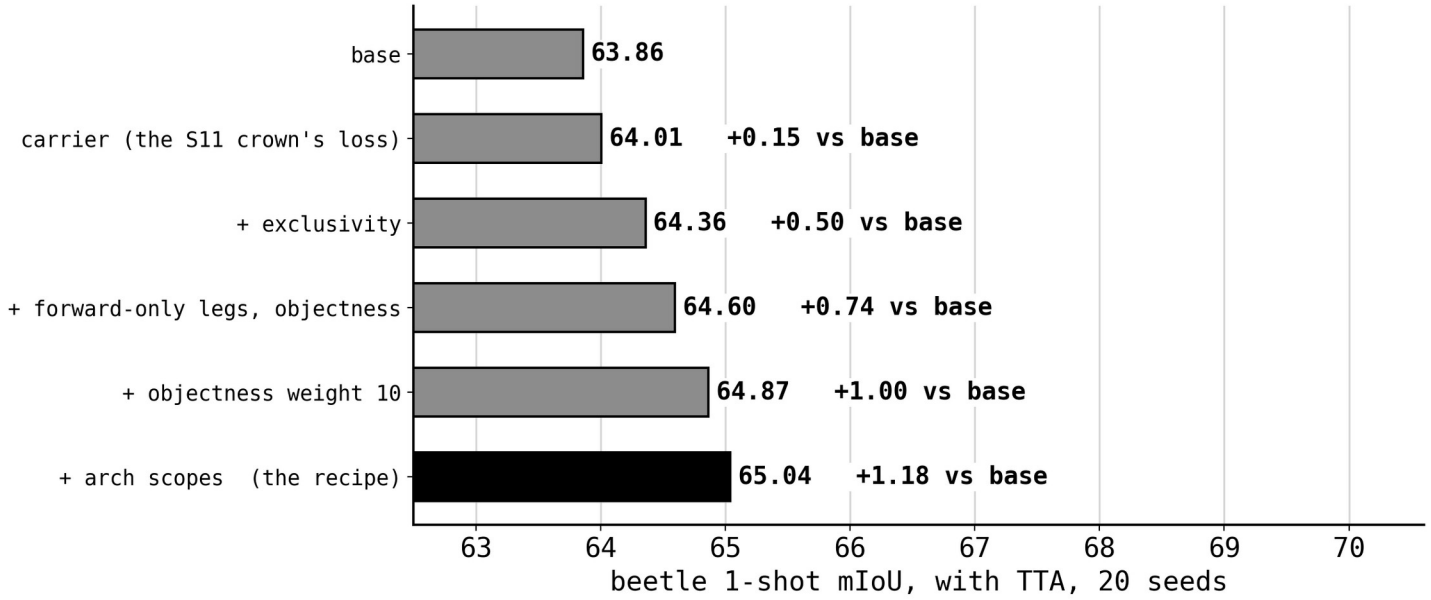
02 The sentinel cuts unfound channels out of the graph. Repair: a term on the forward objectness score, which is upstream of it.

03 Exclusivity, kept from the earlier slate: the one closing-loss change that paid on its own.

All three are inside the fine-tune. No model-code change, no extra labels, no change to the evaluator.

S17B AND S17C

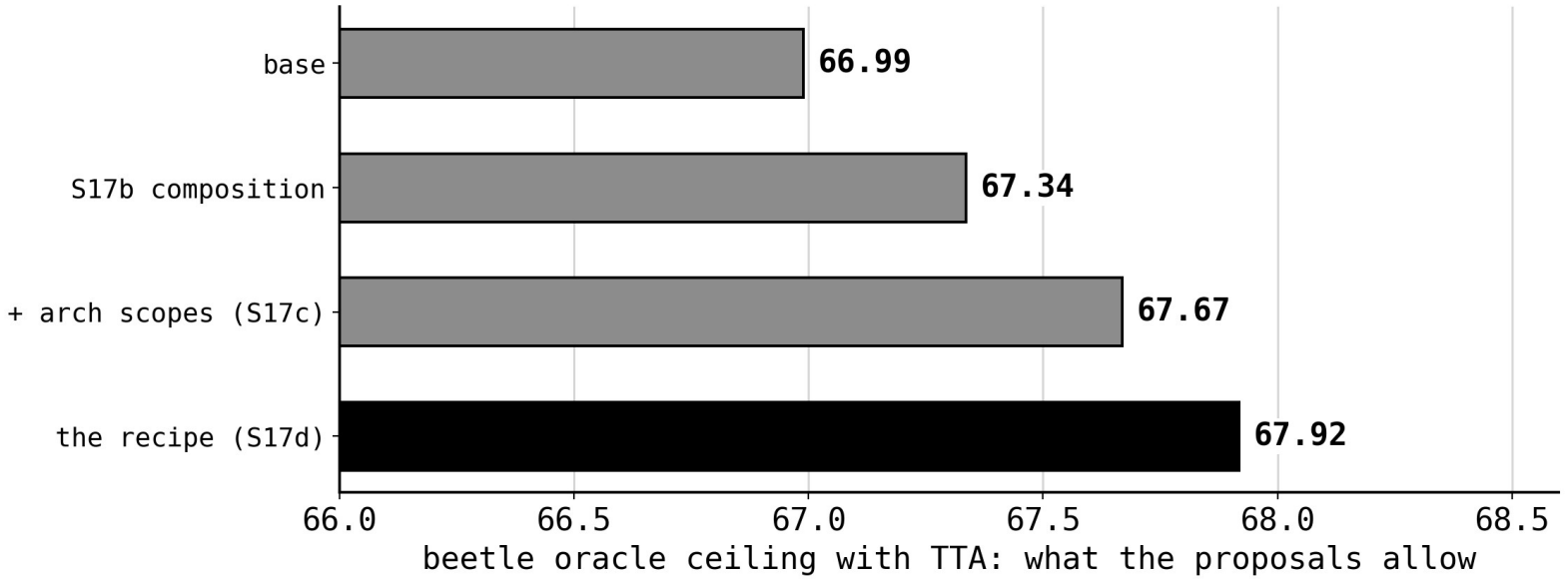
The repairs compose



+0.69 with TTA on 19 of 20 seeds for the composition, the first beetle fine-tuning gain on SAM 2 that is not a reshuffle of channels the model already found.

S17B

The ceiling moves, finally

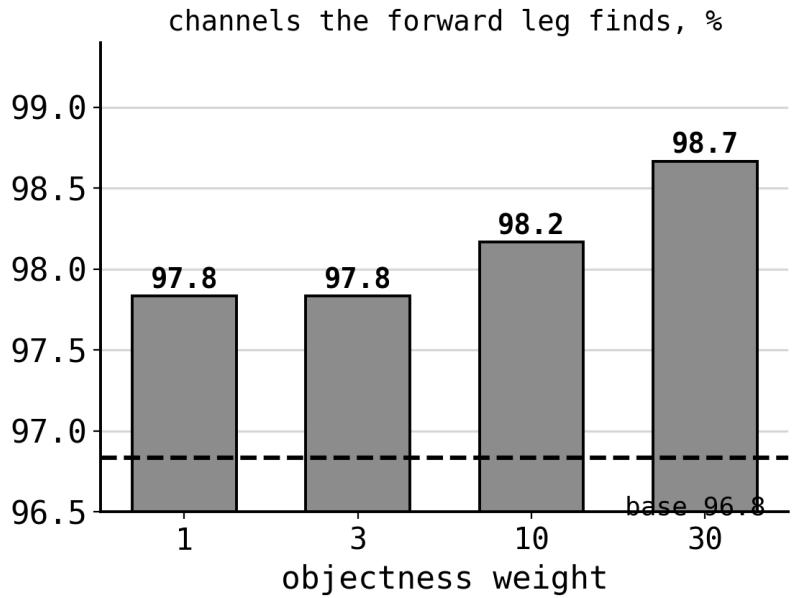


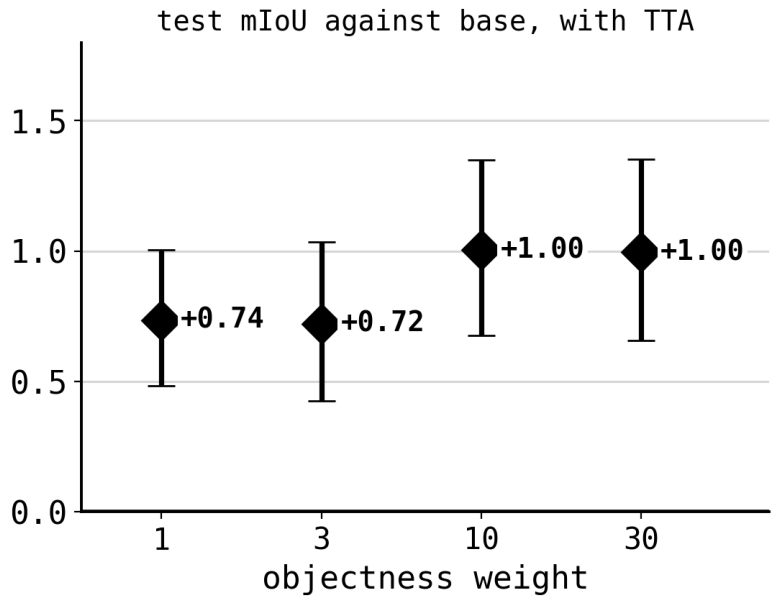
The ceiling is what the proposals allow. On SAM 2 it had not moved for any loss arm in the project.

It moves here, which says the fine-tune changed what gets proposed rather than how the proposals are labelled.

S17C

The prior is free up to 30x



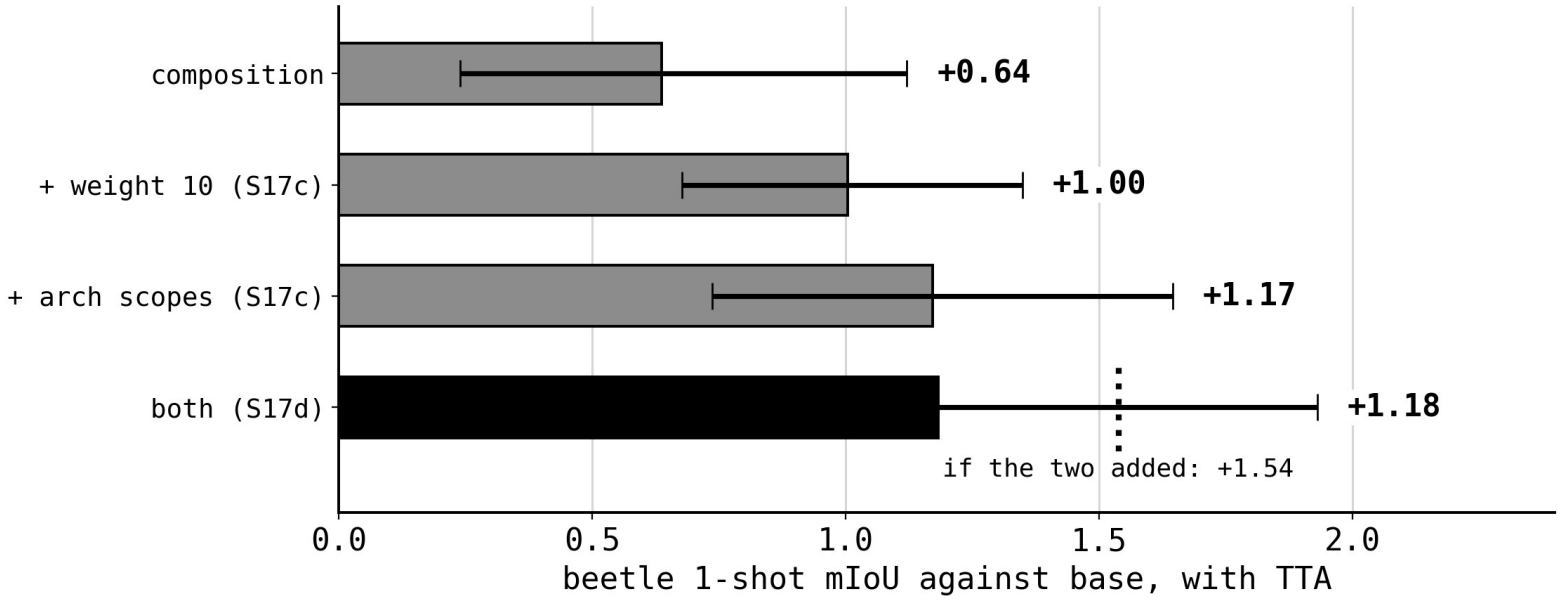


The pre-registration expected a cost at 30. It never appeared: the curve is a plateau, not a peak.

The prior pushes on the objectness score, which is exactly the quantity the sentinel reads.

S17C AND S17D

Scope subsumes the weight

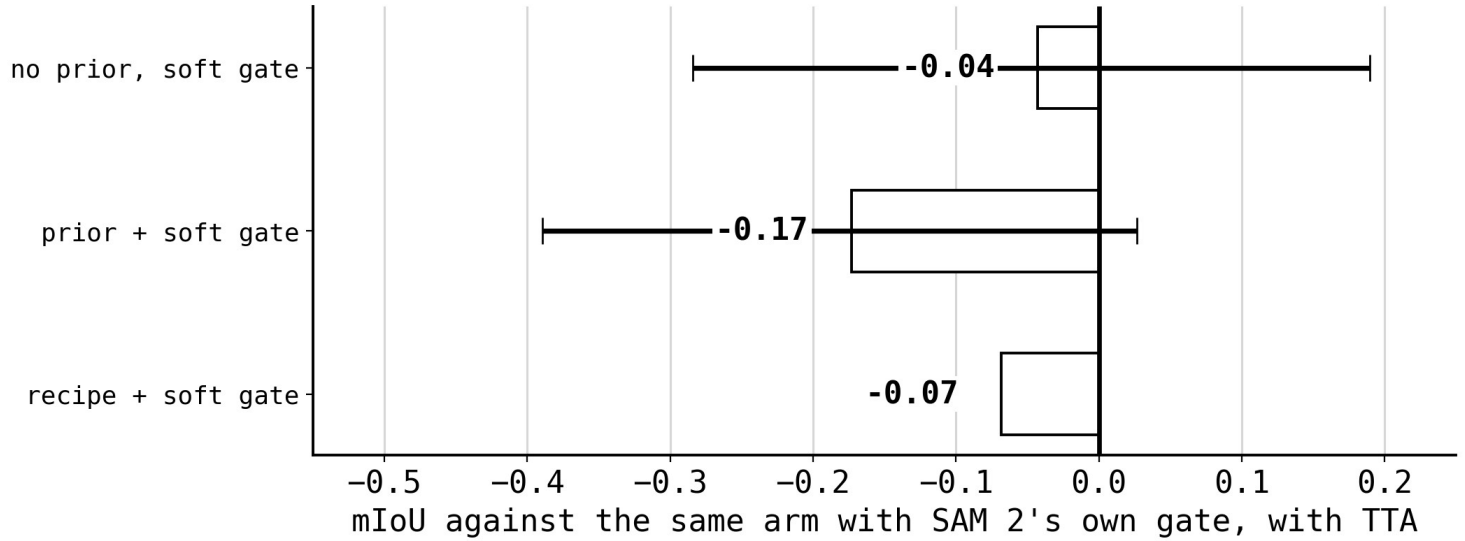


Two levers, each real on its own, and the composition lands below their sum. Pre-registered as either +0.71 if they add or +0.44 if the scope subsumes the weight.

65.04 against 65.03 for the scope alone, +0.01 on the same seeds across two sessions.

S17D

The fence, made differentiable

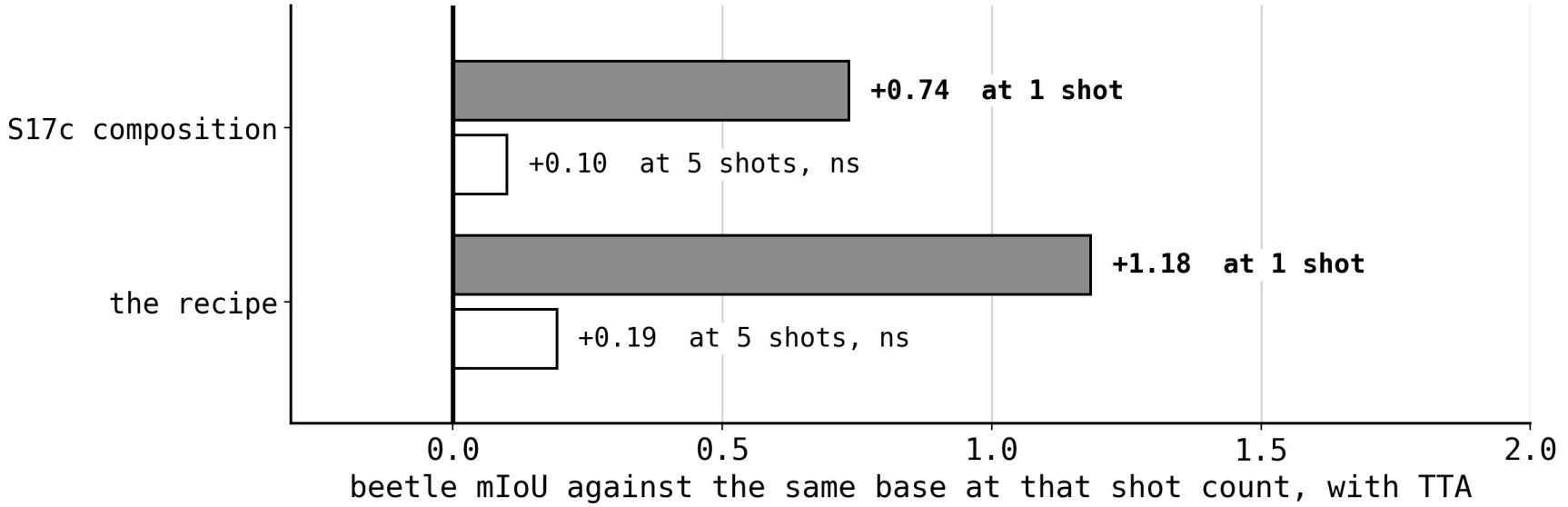


Replacing SAM 2's hard sentinel with a straight-through surrogate is exact at base weights and turns the score head's gradient from 0 into 0.566 on a real pair.

It then flips 1 of 19 fence channels on its own, and costs -0.13 on top of the prior. The loss is now honest about the fence and does not want it.

S17C AND S17D

None of it reaches five shots



Three SAM 2 adapters have now been measured at five shots and none of them moves the number.

Consistent with S15: at five supports a trait absent from one reference is present on another, so the fence the fine-tune repairs is not standing. That reading is not itself measured.

THE OUTPUT

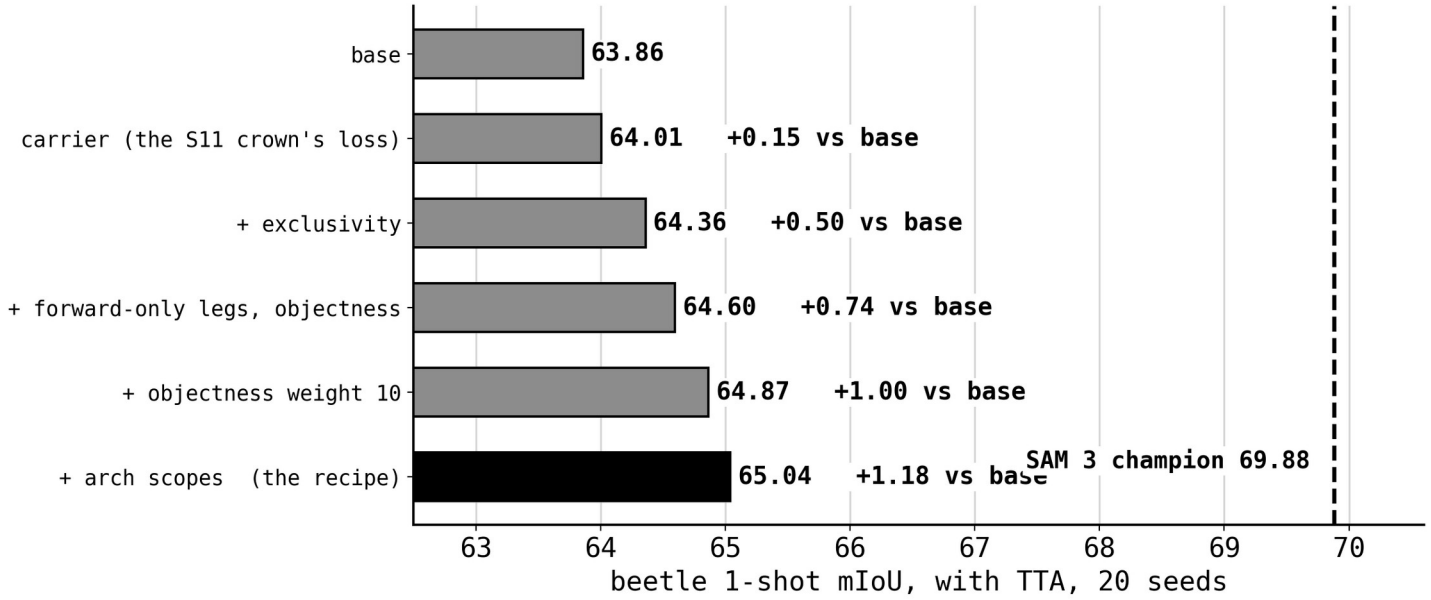
sam2_crown, established

forward-only legs	removes the absorption
objectness prior, weight 10	reaches past the sentinel
exclusivity	the one closing-loss change that paid
arch scopes	puts the proposal-forming module in scope
the S11 carrier	per-trait, lr 1e-5, one epoch, r=32

Pinned in the registry and re-derivable from a record's own fields. Every future SAM 2 fine-tune uses it, and a stage that departs from it says so in its brief.

THE FAMILY IS CLOSED

Where SAM 2 stops

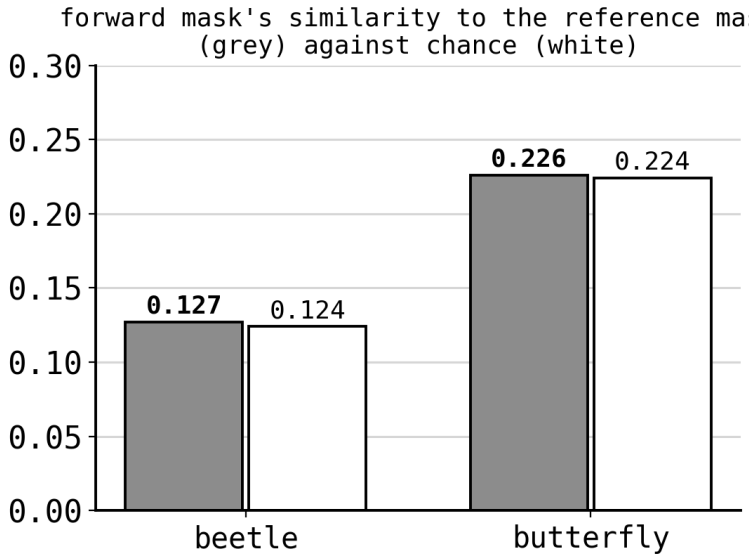


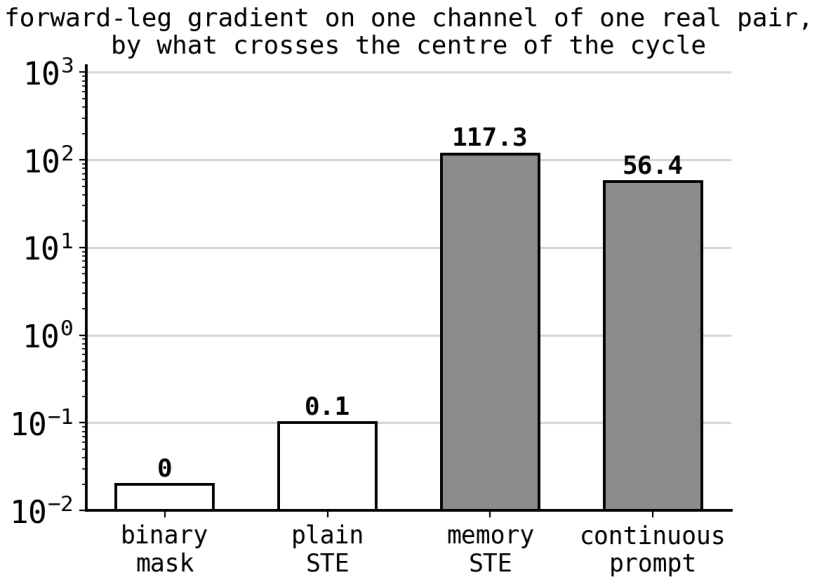
65.04 with TTA, +1.18 over base, oracle ceiling 67.92. That is 0.16 short of the paper's own OC-CCL row and 4.8 below the SAM 3 champion.

Every remaining axis inside the family is measured and closed. What is left is outside it.

NEXT, BUILT AND WAITING

What S18 asks





The cycle is satisfied without the forward mask improving. The mask itself is not being carried: its similarity to the reference sits at chance.

What is left is the prompt. It hands the reverse leg a full continuous logit map, a channel the evaluator never scores. S18 binarises it and keeps the gradient.

AFTER S18

The port that is left

01 Carry the two repairs that worked, forward-only legs and the objectness prior, to SAM 3, where the same memory-attention arm already sits at 69.88.

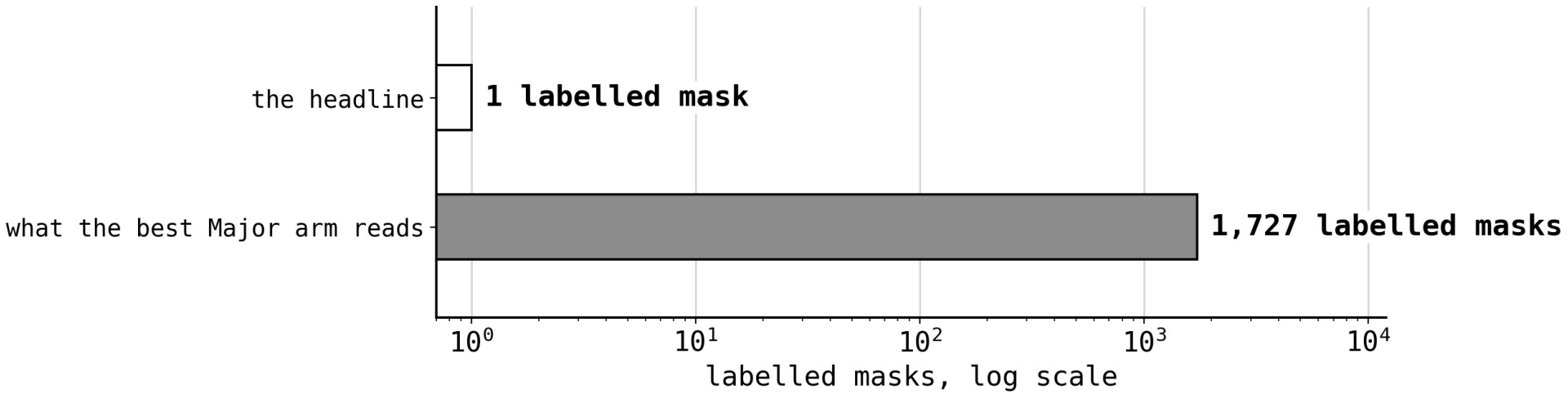
02 The parts are family-agnostic by construction: both were written against per-object logit stacks so they could transfer.

03 The 5-shot result says to expect this at one shot only, and that is a prediction worth writing down before the run.

Cost is one session. The alternative on the same axis is a second support, which S16 already priced at one label.

THE UNCOMFORTABLE AXIS

What the headline hides



SST's claim is about labels. The best Major arm reads 1,727 labelled masks to choose which single reference to use.

The defensible version is that curation stands in for human judgement, and a human does not need 1,727 masks. That is testable offline: shrink the pool and find where the drawn support stops changing.

If it converges at about 20, the headline becomes 20 labels, which is a different paper.

THE OPEN GAP

Major's residual, after all that



Rotation handling is worth 4 to 5 points and is the only lever that ever moved this dataset. Everything else lands at zero.

Two model families at two shot counts leave it where it was, so the remaining candidate is the protocol or the annotations, not capacity.

WHERE IT GOES

Three sentences

01 Beetle is done as a reproduction: it exceeds the paper at one shot and at five, and the label-cheapest routes to it are now priced.

02 SAM 2's objective has been taken apart and put back together, and it tops out 4.8 below what a model swap already gives.

03 The next three moves are S18's mechanism, the port of the two repairs to SAM 3, and the label-budget audit that decides what the headline can say.

Every number in this talk traces to a bundle in results/ and a gate that passed.

BACKUP

The gates behind this talk

S15	champion at 5-shot, both datasets	PASS 18/18	2026-09-13
S16	implicit vs explicit second reference	PASS 14/14	2026-09-01
S17	the cycle's own images, 5 arms	PASS 20/20	2026-09-16
S17b	forward-only legs, objectness	PASS 20/20	2026-09-16
S17c	weight, lr, epochs, scope, link	PASS 11/11	2026-09-17
S17d	scope x weight, the soft sentinel	PASS 11/11	2026-09-17
S18	the palindrome's bottleneck	built, smoke green	pending

Owed, and named in the README: the s11, s13 and s14ft bundles for their formal gates, which sit on Drive rather than in results/inbox.

BACKUP

The cells this talk quotes

beetle 1-shot, SAM 2 base	63.20 plain	**63.86 + TTA**
beetle 1-shot, the recipe	63.71 plain	**65.04 + TTA**
beetle 1-shot, SAM 3 champion		**69.88 + TTA**
beetle 5-shot, SAM 2 base	73.09 plain	**75.16 + TTA**
beetle 5-shot, SAM 3 champion		**78.91 + TTA**
Major 1-shot, best		**76.64 + TTA**
Major 5-shot, SAM 3 base	72.30 plain	**77.95 + TTA**
Major 5-shot, best (SAM 2)		**78.07 + TTA**

20 seeds per cell. Every cell and every paired delta is in the stage bundle it came from.